

Adding context to Lexical Search:

BM25->SPLADE

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Carlos Lassance (now at Cohere)

Based on work with

Benjamin Piwowski (Sorbonne University)

Thibault Formal, Hervé Déjean, Simon Lupart (now at UvA)



NAVER LABS
Europe

Introduction



75% of Web Searches in South Korea

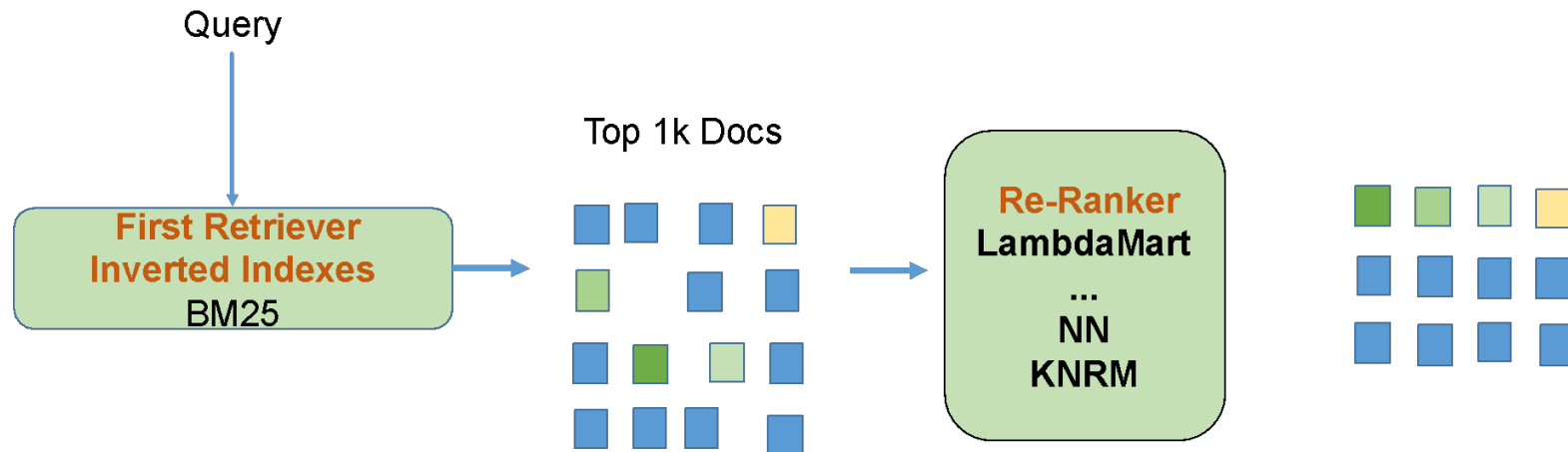
42M Users

Much more than a Search Engine

Outline

- **An Introduction to Neural IR**
- **SPLADE**
- **Demo/Extensions**
 - **Efficiency**

Anatomy of a Search Engine: Thinking Fast and Slow



BM25, Robertson et al., 1994

- Hypothesis: word frequencies follow a two Poisson Mixture

$$\sum_{w \text{ in } q^d} \frac{tf(w)}{tf(w) + K} IDF(w)$$

- The backbone of search engines for several decades

Classical Rerankers

Rerankers: Learning-to-rank methods :

- LambdaMart, RankNET, GBDT on handcrafted features

2010's: NN models with word embedding (word2vec)

- **Representation based** e.g. DSSM
- **Interaction based** e.g. DRMM, K-NRM, DUET

Longstanding debate in Information Retrieval

DENSE

$\mathbb{R}_{\text{hundreds}}$

Semantic Match
Representation
Based
Approximate NN

DENSE



SPARSE

SPARSE

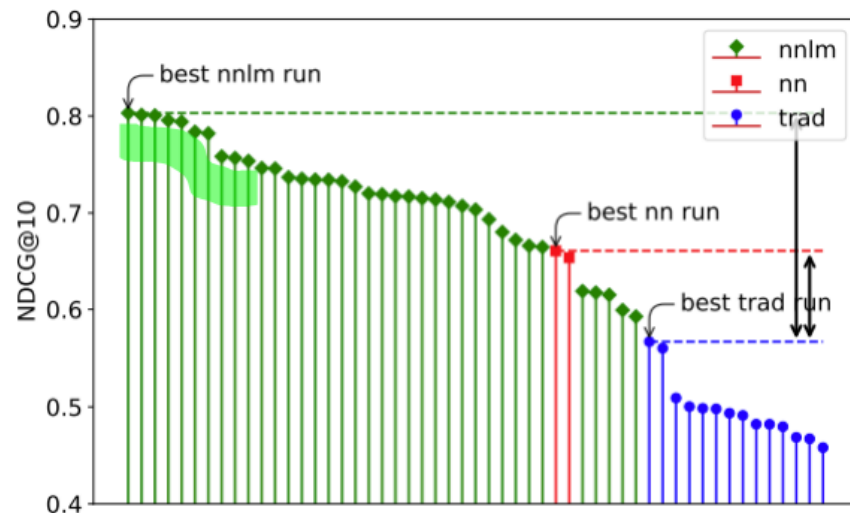
$\mathbb{R}_{100k-M}$

Exact Match
Interaction Based
Inverted Index

MSMARCO and TREC

Information Retrieval Competition since 90's

2019



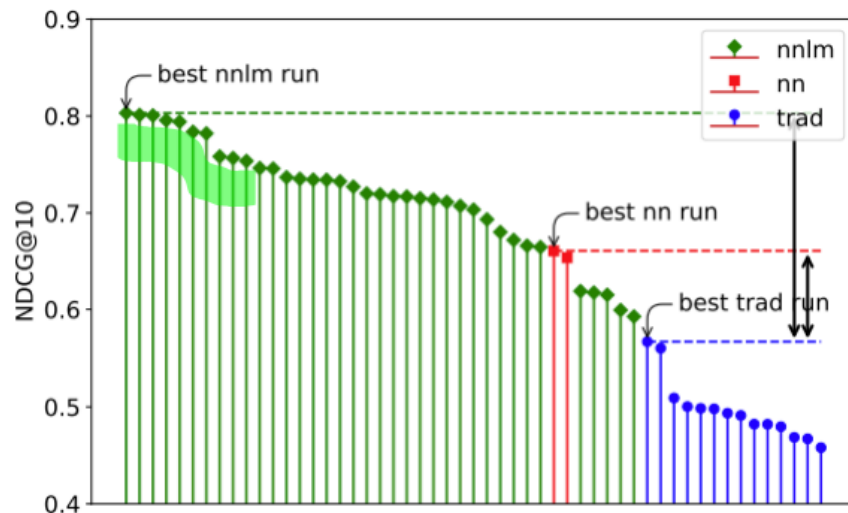
MSMARCO and TREC

Information Retrieval Competition since 90's

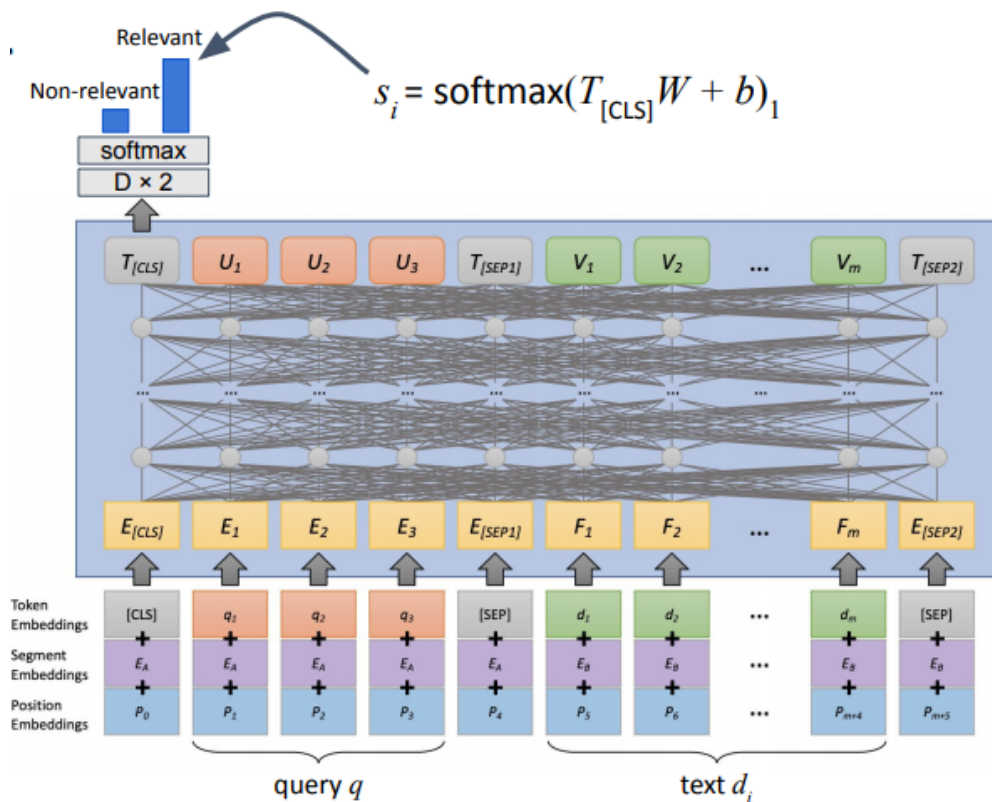
2019



Huge Gain but High Computational Cost



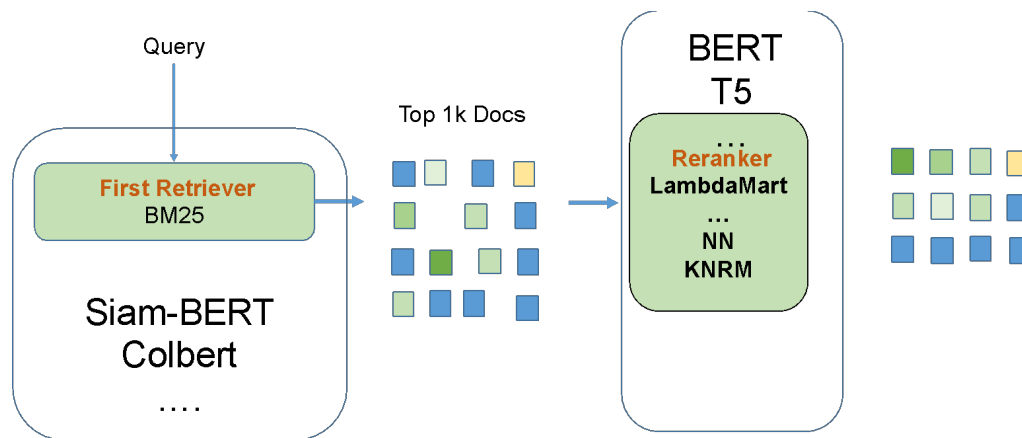
BERT Reranker: Cross-Encoders



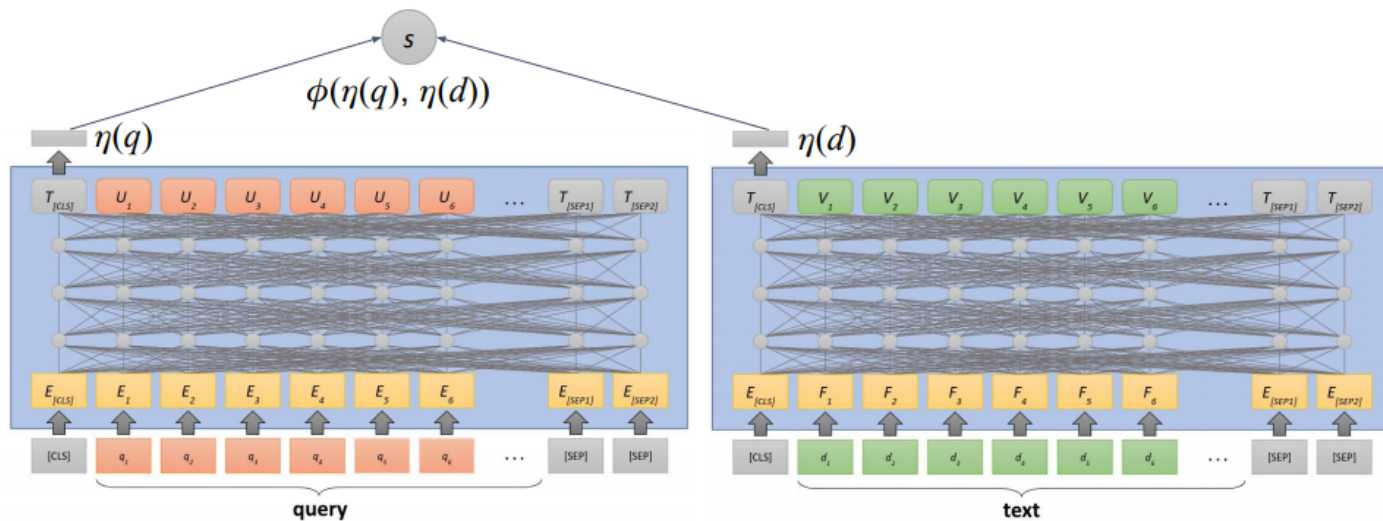
FT with various learning to rank loss **on the Top1k documents returned by BM25**

Schema credit: Lin Nogueira, Yates in **Pretrained Transformers for Text Ranking: BERT and Beyond**

Pretrained LMs for First Retriever and Rerankers



A Bi-Encoder First Stage Ranker



• From Inverted index to dense indexing technique (ANN)

First Ranker Comparison: MS-Marco and TRECDL'19

Model	ndcg@10
BM25	50.1
Bi-Encoder (DPR)	63.7
TAS-B	71.7

+ cross encoders Distillation,
negative mining etc

BEIR Benchmark: Zero Shot Evaluation, Neurips'21



BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, Iryna Gurevych

Ubiquitous Knowledge Processing Lab (UKP-TUDA)

Department of Computer Science, Technische Universität Darmstadt

www.ukp.tu-darmstadt.de



Figure 1: An overview of the diverse tasks and datasets present in BEIR.

BEIR Conclusion: BM25 > Dense Bi-Encoders

“Our results **show BM25 is a robust baseline** ... In contrast, Dense-retrieval models [...] often underperform other approaches, highlighting the considerable room for improvement in their generalization capabilities .

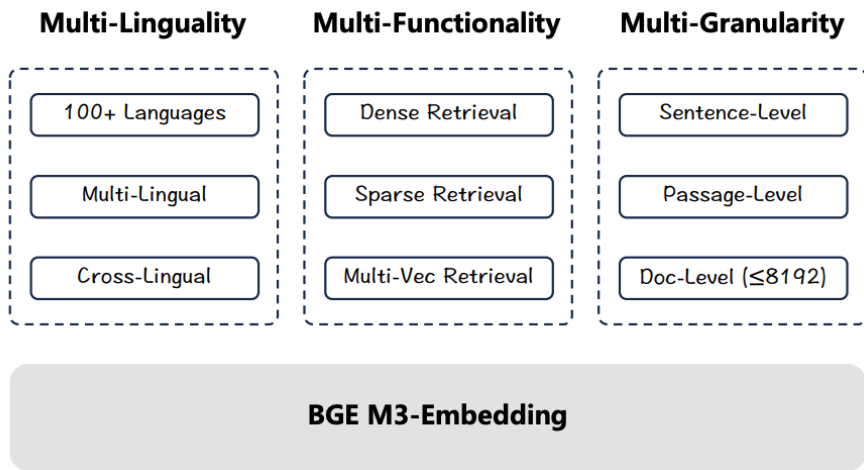
Mean ndcg@10 over 18 Datasets

BM25	TAS-B
45.3	43.7

- Rerankers transfer well
- Standard bi-encoders don't

More recently on BM25

AI/ML people do not know how to use BM25 (cf BGE, Monarch)



BM25 from Paper	BGE-M3	BM25 @Carlos
28.9	65	64.1

Longstanding debate in Information Retrieval

DENSE

$\mathbb{R}_{\text{hundreds}}$

Semantic Match
Representation
Based
Approximate NN

DENSE



SPARSE

$\mathbb{R}_{100k-M}$

Exact Match
Interaction Based
Inverted Index

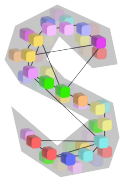
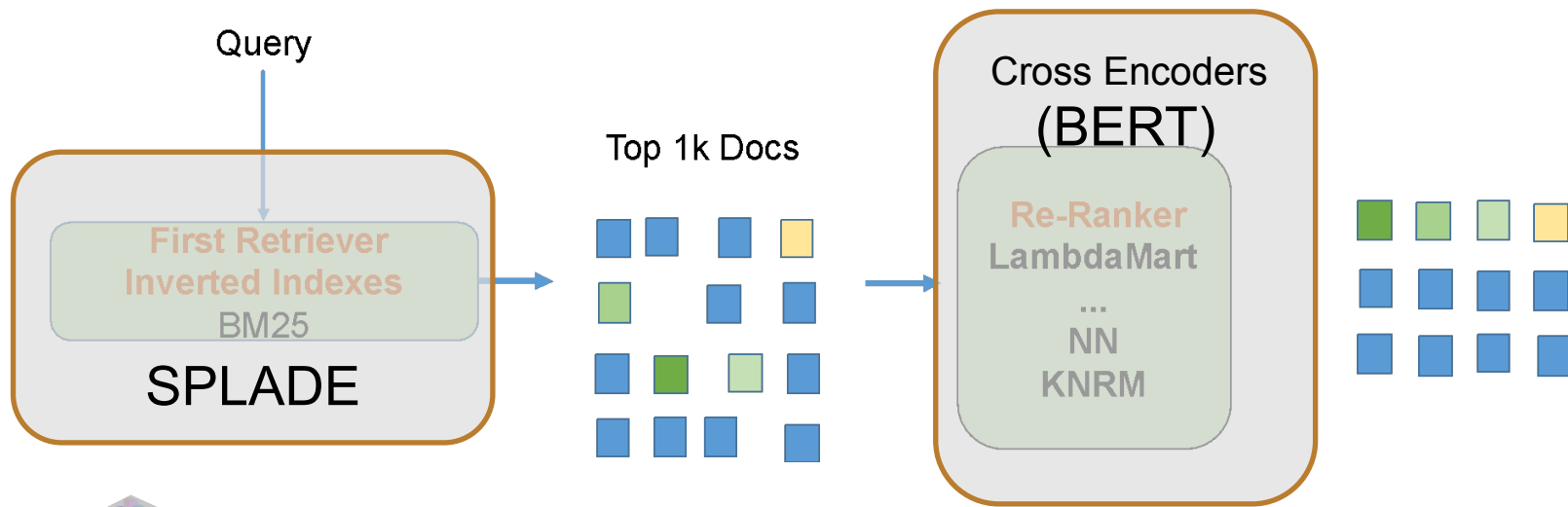
Can we learn a sparse model?

SPLADE

NA ER



SPLADE: An Efficient and Effective Sparse Ranker



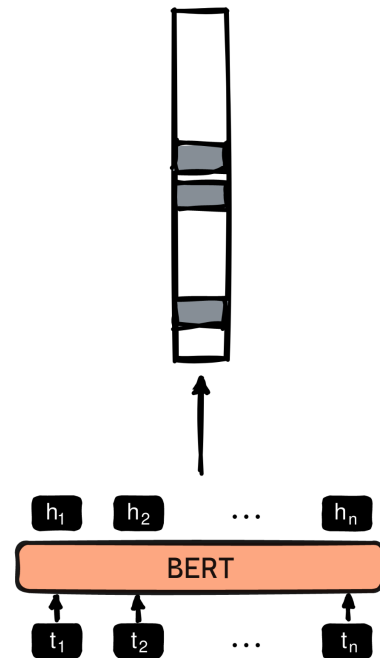
RQ How to learn state-of-the-art sparse retrieval models?

Learned Sparse Model

Represent queries and documents as high-dimensional sparse vectors

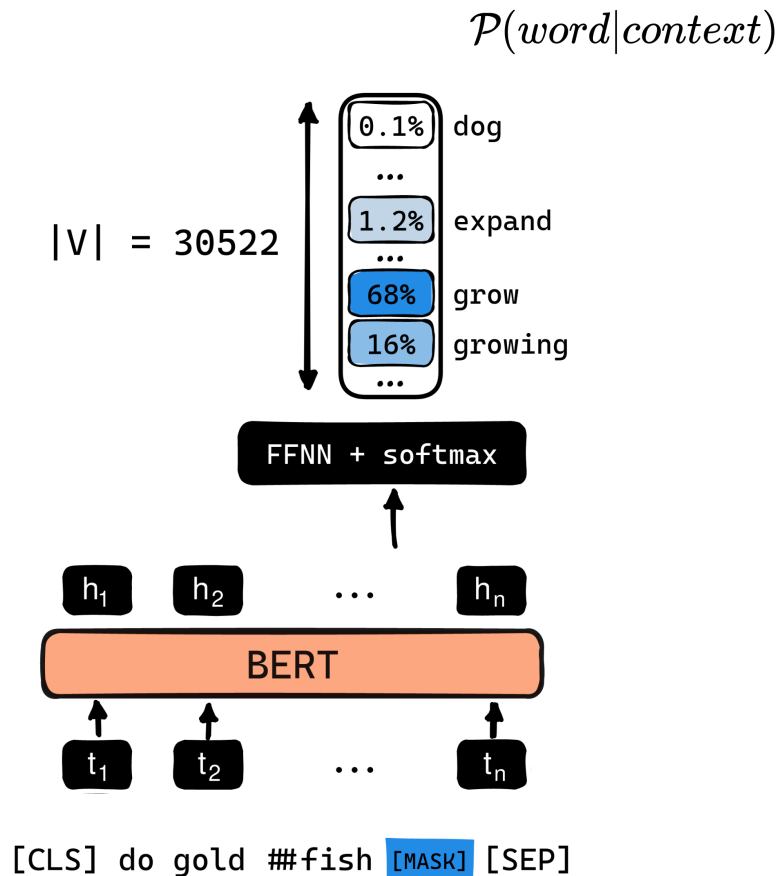
RQ Can we design a sparse model that is

- Effective
- Efficient
- Interpretable
- Robust



The MLM Head

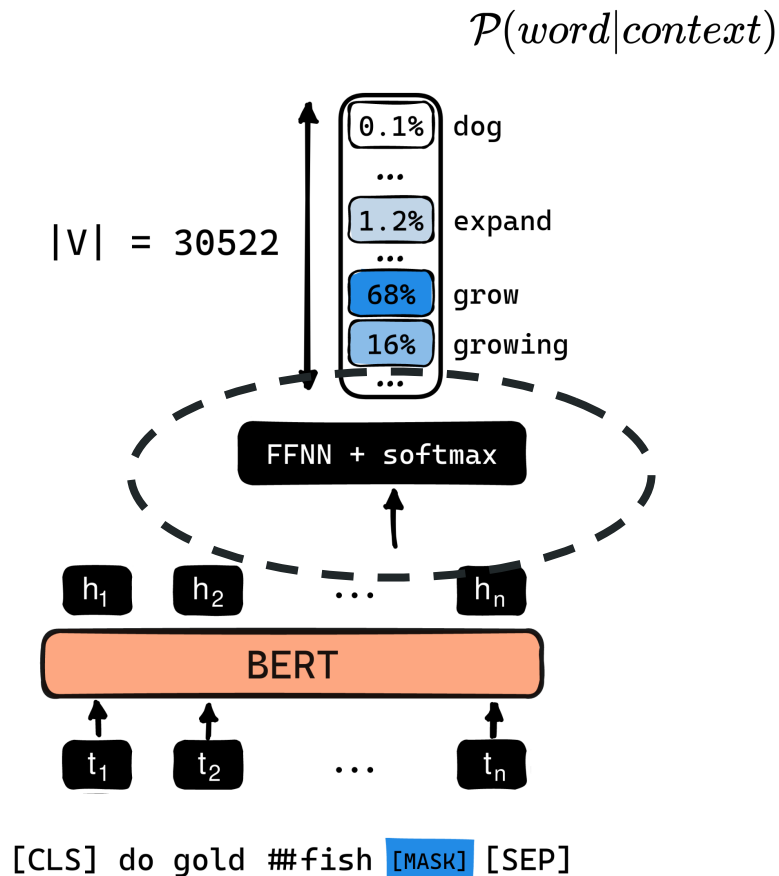
BERT implicitly learns to perform expansion



The MLM Head

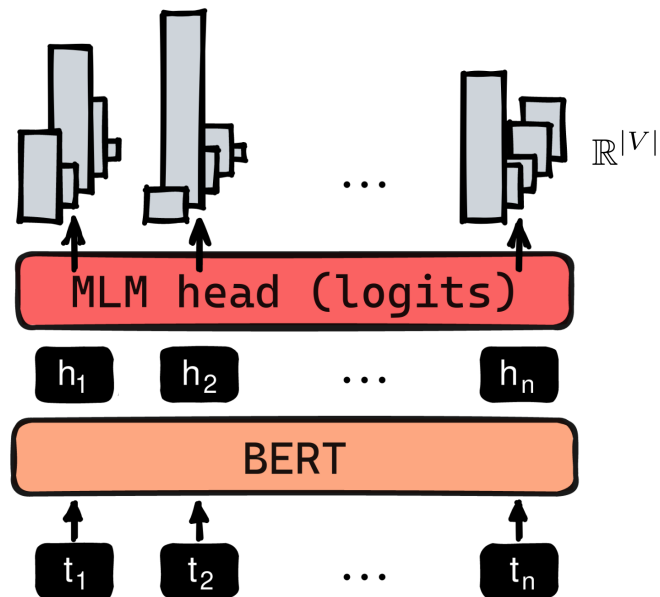
BERT implicitly learns to perform expansion

Idea re-use the MLM head instead
of throwing it away!



SPLADE Architecture

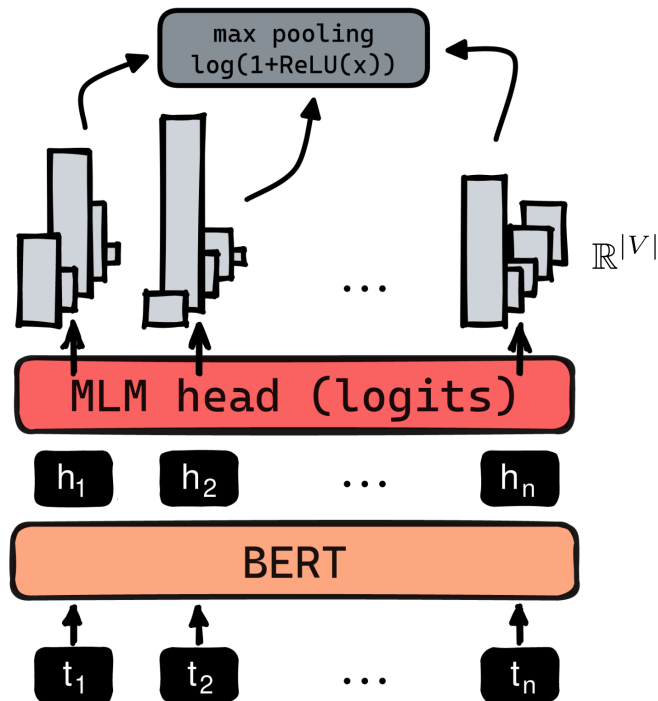
Re-use the MLM head



SPLADE Architecture

Log saturation on term weights: prevents some terms to dominate

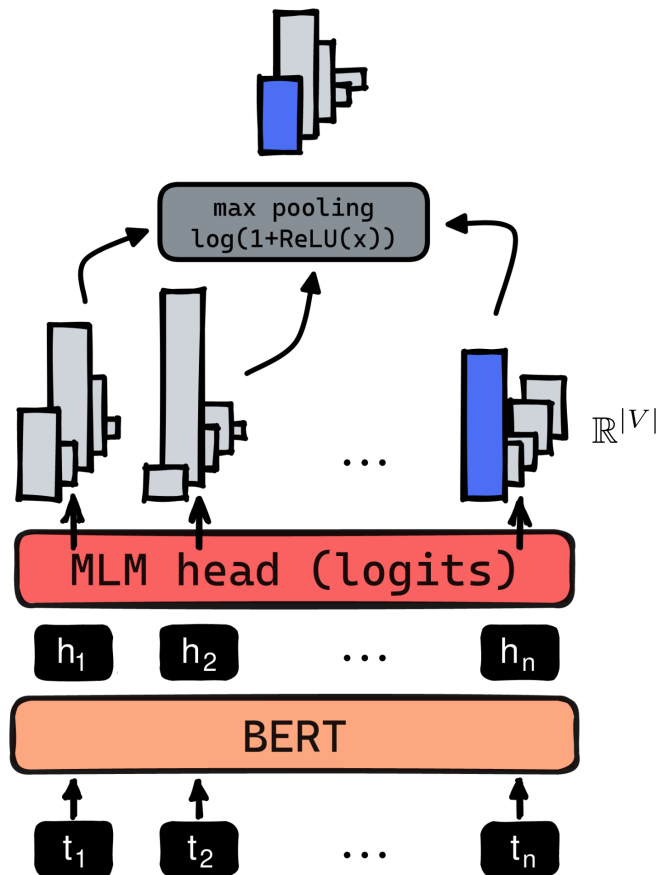
- Traditional $\log(tf)$ [Manning *et al.* 2008]
- EPIC [MacAvaney *et al.* 2020]
- SPARTA [Zhao *et al.* 2021]



SPLADE Architecture

Log saturation on term weights: prevents some terms to dominate

Max pooling [Formal *et al.* 2021a]



SPLADE Architecture

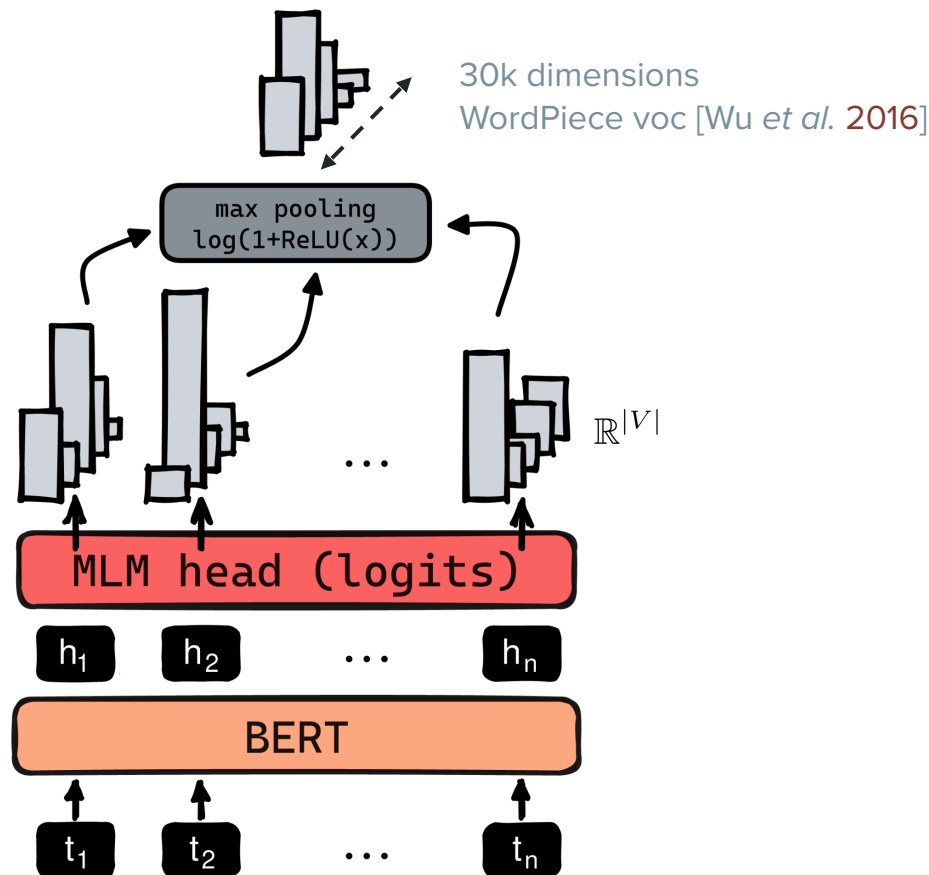
Log saturation on term weights: prevents some terms to dominate

Max pooling [Formal *et al.* 2021a]

The model both learns

- Term **expansion**
- Term **weighting**

$$s(q, d) = \phi(q)^T \phi(d)$$



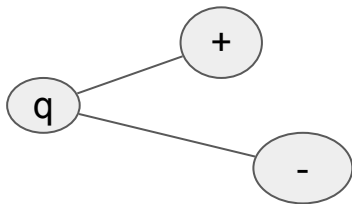
An example

original document (doc ID: 7131647)	
if (1.2) bow (2.56) legs (1.18) is caused (1.29) by (0.47) the bone (1.2) alignment (1.88) issue (0.87) than you may be able (0.29) to correct (1.37) through (0.43) bow legs correction (1.05) exereises. read more here. if bow legs is caused by the bone alignment issue than you may be able to correct through bow legs correction exercises.	
	stemming effect
bad expansion terms ! expansion terms	
(leg, 1.62)	(arrow, 0.7)
(exercise, 0.64)	(bones, 0.63)
(problem, 0.41)	(treatment, 0.35)
(happen, 0.29)	(create, 0.22)
(can, 0.14)	(worse, 0.14)
(effect, 0.08)	(teeth, 0.06)
(remove, 0.03)	

good expansion terms

SPLADE Training

- Contrastive Loss (InfoNCE) or Distillation Loss from a Cross-Encoders



- Sparsity loss to optimize *the retrieval FLOPS* (with an inverted index)
 - Prevent too frequent term expansion (FLOPS regularization != L1 loss)

SPLADE Achievements

- State of the art Sparse Retrieval Model in 2022
- State of the art on the BEIR benchmark in 2022! Generalize well!
 - Mean ndcg@10

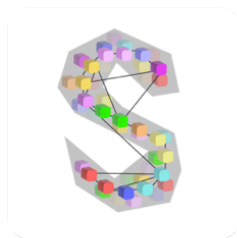
BM25	TAS-B	SPLADE'22
44.0	43.5	50.7

- Splade Popularity:
 - 🧡 >Ms 'downloads'
 - Revive Academic Interest for Sparse Models

SPLADE in 2024

- Where can I use it?
- Sample applications
- How efficient it really is?

SPLADE adoption in 2024 (Explicit mention)

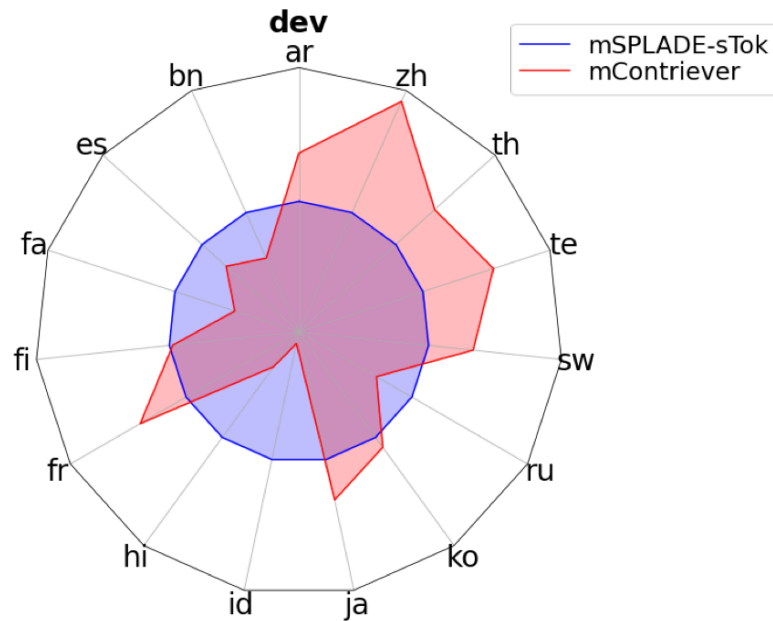


TREC tasks

- Annual conference with different IR tasks (8 in 2024)
- Annotation based on participants
 - No “winner”, but... “highly performant solution” (HPF)
- In 2023
 - SPLADE part of HPF in 6 out of 8
 - Product-search, ClinicalTrials, MultiModal, MultiLingual and “Deep Learning”
 - In 3 out of 6, NLE had no involvement

Multilingual SPLADE

- We have been able to make it work well
- Over 18 languages
- Hard to train
 - Multilingual -> More tokens
- Won the WSDM Cup 2023
 - Part of the solution, not complete one



MultiModal SPLADE

- Two implementations
 - STAIR – Chen et al 2023 -> APPLE
 - Unsupervised over private large dataset
 - LexLIP – Luo et al 2023 -> Microsoft
 - Supervised over public data
- We tried our own
 - Inconclusive



airport: 1.6, airline: 1.35, airlines: 1.22, window: 1.22, airports: 1.21, delta: 1.21, nexus: 1.06, lax: 0.99, concourse: 0.97, ##fk: 0.92, gate: 0.87, flights: 0.86, traveled: 0.84, view: 0.82, airways: 0.79, aircraft: 0.76, mildly: 0.76, ##lr: 0.75, rainy: 0.75, traveling: 0.75

Efficiency

- No free-lunch
- State of the art
- Two-Step SPLADE

Efficiency - No free-lunch

Findings:

- Recent sparse models are slower than BM25
- SPLADE 50x slower on mono-thread evaluation

Method	Quality	Time	Space
	RR@10	Latency (ms)	Index Size (MB)
Anserini (Lucene): DAAT			
(1a) BM25	0.187	40.1	661
(1b) BM25-T5	0.277	62.8	1036
(1c) DeepImpact	0.325	244.1	1417
(1d) uniCOIL-T5	0.352	222.3	1313
(1e) uniCOIL-TILDE	0.350	194.6	2067
(1f) SPLADEv2	0.369	2140.0	4987

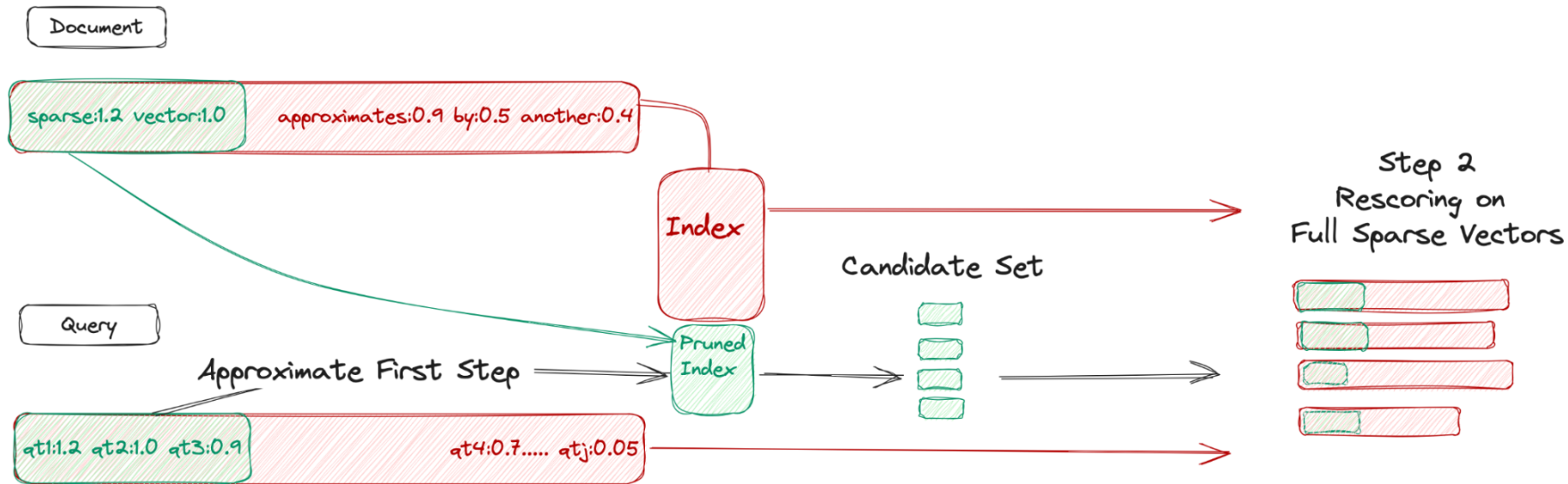
Efficiency - State of the art

- Completely redesign training for efficiency
 - Efficient-SPLADE (not good OOD) -> <https://arxiv.org/pdf/2207.03834.pdf>
- Approximate Sparse retrieval
 - Pruning (SIGIR 2023) -> <https://arxiv.org/pdf/2304.12702.pdf>
 - **TwoStep** (following slides, ECIR2024, ELASTICSEARCH)
 - Convert to dense, use dense techniques (PINECONE, Zilliz, Benchmark NeuriPS)
- Modify the inverted index algorithm
 - Guided Traversal (Mallia et al) -> <https://arxiv.org/abs/2204.11314>
 - Postings Clippings (Mackenzie et al) -> <https://jmmackenzie.io/pdf/mmmp22-emnlp.pdf>

Efficiency - Two-Step SPLADE

- What can we do with a fixed model?
 - No retraining
- Pruning can be the solution
 - Indexes have redundant info
- Take lessons from dense
 - Approximate search + Rescoring
- Elasticsearch went with this option concurrently to us
 - Around october of last year
 - <https://www.elastic.co/search-labs/blog/articles/introducing-elser-v2-part-2>

Efficiency - Two-Step SPLADE



Efficiency - Two-Step SPLADE (ElasticSearch)

```
GET /my-index/_search
```

```
{  
  "query": {  
    "bool": {  
      "should": [  
        {"term": {"tokens": {"value": "<kept token 1>", "boost": <kept weight 1>}}},  
        {"term": {"tokens": {"value": "<kept token 2>", "boost": <kept weight 2>}}},  
        ...  
      ]  
    }  
  },
```

non-frequent or critical tokens

```
    "rescore": {  
      "window_size": 50,  
      "query": {  
        "rescore query": {  
          "bool": {  
            "should": [  
              {"term": {"tokens": {"value": "<dropped token 1>", "boost": <dropped weight 1>}}},  
              {"term": {"tokens": {"value": "<dropped token 2>", "boost": <dropped weight 2>}}},  
              ...  
            ]  
          }  
        }  
      },
```

frequent and non-critical tokens

```
      "query_weight" : 1,  
      "rescore_query_weight" : 1  
    }  
  }  
}
```

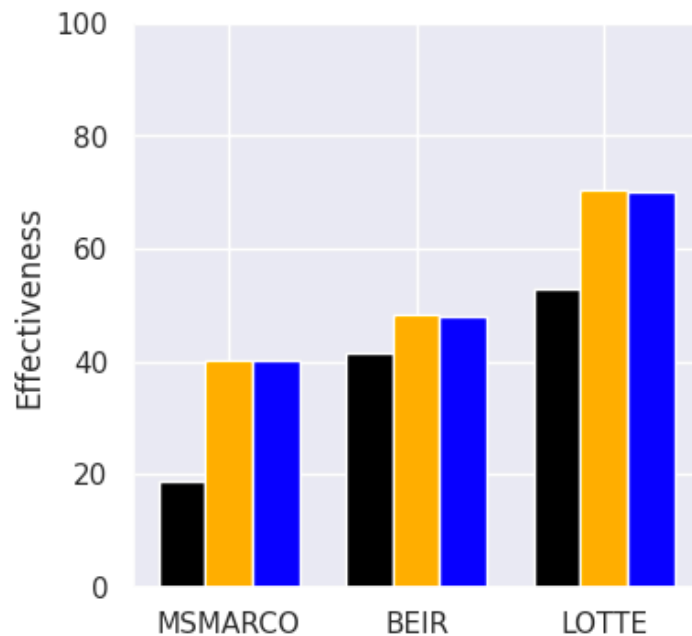
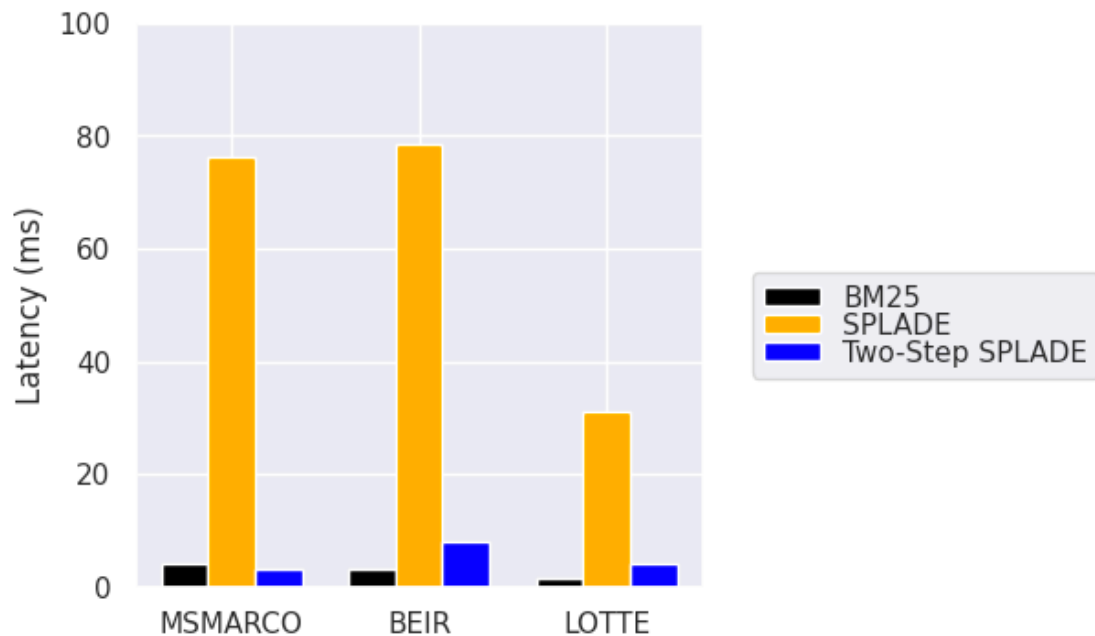
$$\text{score} = 1 \times (\text{"query kept token 1 weight"} \times \text{"doc kept token 1 weight"} +$$

$$\text{"query kept token 2 weight"} \times \text{"doc kept token 2 weight"} + \dots) +$$

$$1 \times (\text{"query dropped token 1 weight"} \times \text{"doc dropped token 1 weight"} +$$

$$\text{"query dropped token 2 weight"} \times \text{"doc dropped token 2 weight"} + \dots)$$

Efficiency - Two-Step SPLADE



Conclusion

- SPLADE is mature after 3 years of research:
 - ~300k downloads last month, 3M in total for SPLADE++ PPED (huggingface)
 - Able to get latency near BM25
 - One of the best models for zero-shot OOD
- Pain-points:
 - We still use tokens and not full words
 - Stemming effects are lost, huge concentration, ambiguity
 - Indexing is much more costly than BM25
 - Training is better, but still too many hyperparameters
 - Impossible to guarantee effectiveness OOD

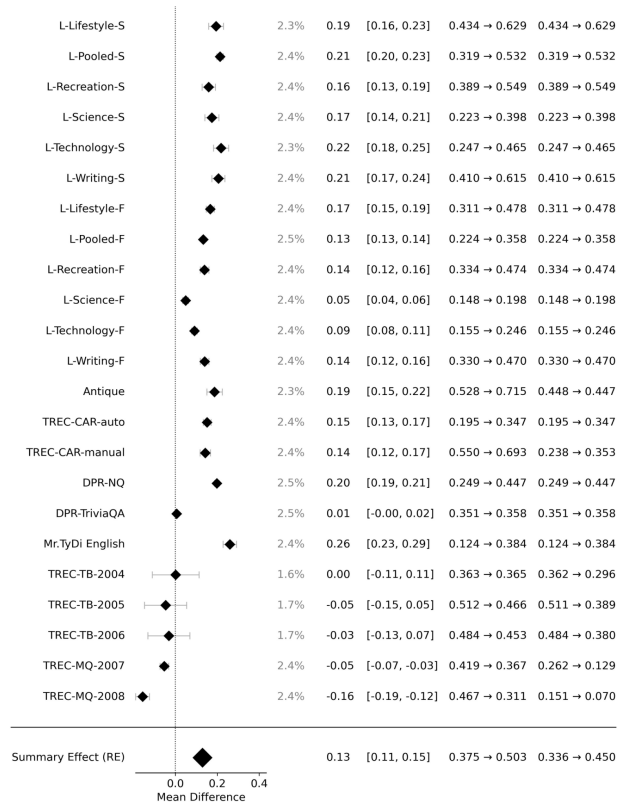
github.com/naver

<https://huggingface.co/naver/splade-cocondenser-ensembledistil>

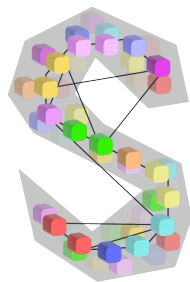
Effectiveness OOD

BM25 vs SPLADE-2023

	Effect Size	Weight	Mean	CI 95%	judged only	All
M-DEV	◆	2.4%	0.25	[0.23, 0.26]	0.223 → 0.469	0.223 → 0.469
M-TREC_2019	◆	1.6%	0.26	[0.15, 0.37]	0.496 → 0.754	0.483 → 0.731
M-TREC_2020	◆	2.0%	0.28	[0.20, 0.36]	0.480 → 0.758	0.476 → 0.756
M-TREC-Hard	◆	1.9%	0.20	[0.12, 0.29]	0.565 → 0.769	0.258 → 0.367
M-TREC_2021	◆	2.0%	0.27	[0.20, 0.35]	0.445 → 0.718	0.427 → 0.703
M-TREC_2022	◆	2.1%	0.37	[0.30, 0.43]	0.281 → 0.650	0.264 → 0.634
M-V2_DEV	◆	2.4%	0.13	[0.11, 0.14]	0.076 → 0.202	0.076 → 0.202
M-V2_DEV2	◆	2.4%	0.13	[0.12, 0.14]	0.082 → 0.213	0.082 → 0.213
B-arguana	◆	2.4%	0.17	[0.14, 0.19]	0.348 → 0.514	0.348 → 0.514
B-climate-fever	◆	2.4%	0.04	[0.02, 0.06]	0.169 → 0.208	0.169 → 0.208
B-dbpedia	◆	2.4%	0.17	[0.14, 0.20]	0.403 → 0.573	0.325 → 0.455
B-fever	◆	2.4%	0.16	[0.15, 0.17]	0.622 → 0.784	0.622 → 0.784
B-fiqa	◆	2.3%	0.13	[0.10, 0.17]	0.240 → 0.373	0.240 → 0.373
B-hotpotqa	◆	2.5%	0.06	[0.06, 0.07]	0.625 → 0.690	0.625 → 0.690
B-nfcorpus	◆	2.2%	0.03	[-0.02, 0.08]	0.328 → 0.358	0.328 → 0.358
B-nq	◆	2.4%	0.28	[0.26, 0.30]	0.306 → 0.586	0.306 → 0.586
B-quora	◆	2.5%	0.10	[0.09, 0.11]	0.709 → 0.813	0.709 → 0.813
B-scidocs	◆	2.4%	0.01	[-0.01, 0.02]	0.152 → 0.158	0.152 → 0.158
B-scifact	◆	2.1%	0.03	[-0.03, 0.09]	0.678 → 0.711	0.678 → 0.711
B-TREC_Covid	◆	1.8%	0.13	[0.03, 0.22]	0.648 → 0.774	0.639 → 0.749
B-Touche	◆	2.1%	-0.08	[-0.15, -0.01]	0.835 → 0.756	0.437 → 0.307



Beers?



github.com/naver

<https://huggingface.co/naver/splade-cocondenser-ensembledistil>